APPENDIX A

Further Topics: Polynomials, Adjoints, and Unitary Diagonalization

MOTIVATION AND GUIDING IDEA

Gram-Schmidt replaces a list of directions with an orthonormal list spanning the same space, simplifying coordinate and projection computations. Orthogonal diagonalization imposes an additional requirement: the basis vectors must also be eigenvectors. For diagonalization, therefore, begin with the eigenspaces and apply the algorithm within each one; applying it to an arbitrary basis does not generally produce eigenvectors.

NOTATION AND CONVENTIONS

- v_j is an input vector, w_j is the residual after subtracting projections, and e_i is a unit vector already obtained. Never divide by $\|w_j\|$ when $w_j = 0$.
- $\text{span}\{v_1, \ldots, v_j\}$ is the span of the input prefix; it equals the span of the output vectors obtained up to that step.
- $\mathbb{R}_n[x]$ and $\mathbb{C}_n[x]$ denote polynomials of degree at most n, including the zero polynomial. Evaluation values and integrals are computed according to the inner product specified in the question.
- $E_\lambda(T) = \ker(T - \lambda I)$ is the eigenspace belonging to λ; an eigenvector is a **nonzero** element of this space.
- T^* is the adjoint map; $A^* = \overline{A}^T$ is the conjugate transpose. The identity $[T^*]_B = [T]_B^*$ requires an orthonormal basis B.
- $T = T^*$ means self-adjointness; $TT^* = T^*T$ means normality.
- $Q^TQ = QQ^T = I$ describes a real orthogonal matrix; $U^*U = UU^* = I$ describes a complex unitary matrix. In particular, $Q^{-1} = Q^T$ and $U^{-1} = U^*$.
- In $Q^TAQ = D$ or $U^*AU = D$, column j of the change-of-basis matrix is an eigenvector belonging to d_{jj}. The column order must match the order of the eigenvalues on the diagonal.

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DEFINITION A.1

Conditions for the tools. In an inner product space over $\mathbb{F} \in \{\mathbb{R}, \mathbb{C}\}$, Gram-Schmidt applies to any finite list, including a dependent list: after obtaining $e_1, \ldots, e_r$, compute

$$w_j = v_j - \sum_{i=1}^{r} \langle v_j, e_i \rangle e_i.$$

If $w_j = 0$, skip that input vector; otherwise append $e_{r+1} = w_j / \|w_j\|$. The resulting list is an orthonormal basis of the input span. For a linearly independent input, no vector is skipped. All projection coefficients use the convention of linearity in the first argument.
If V is finite dimensional and $T : V \to V$ is linear, there is a unique linear map $T^* : V \to V$ satisfying $\langle Tv, w \rangle = \langle v, T^* w \rangle$ for all v, w. Over $\mathbb{C}$, a map is normal if and only if it has an orthonormal eigenbasis. If it is self-adjoint, all its eigenvalues are real. Over $\mathbb{R}$, every symmetric matrix admits an orthogonal diagonalization; normality alone does not suffice for a real orthogonal diagonalization. For example,

$$R = \begin{pmatrix} 0 & -2 \\ 2 & 0 \end{pmatrix}, \qquad RR^T = R^T R = 4I,$$

but $R^2 = -4I$, so R has no real eigenvector. When constructing an orthonormal eigenbasis, apply Gram-Schmidt **within each eigenspace** so that the linear combinations remain in that eigenspace.

A.1 Adjoints, Normality, and Diagonalization in an Orthonormal Basis

DEFINITION A.2

Let V be a finite-dimensional inner product space over $\mathbb{F} \in \{\mathbb{R}, \mathbb{C}\}$, and let $T : V \to V$ be linear. The **adjoint** T^* is the unique linear map satisfying

$$\langle Tv, w \rangle = \langle v, T^* w \rangle \qquad \text{for all } v, w \in V.$$

In an orthonormal basis, if A represents T, then $A^* = \overline{A}^T$ represents T^*. We call T **self-adjoint** if $T^* = T$, and **normal** if $TT^* = T^*T$. A matrix $U \in M_n(\mathbb{C})$ is **unitary** if $U^*U = UU^* = I$; its columns form an orthonormal basis of $\mathbb{C}^n$. Over the reals, a square matrix Q is **orthogonal** if $Q^TQ = QQ^T = I$.

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A.2 Polynomials over the Complex Numbers

DEFINITION A.3

The notation $\mathbb{C}[X]$ denotes the polynomials in the variable X with complex coefficients:

$$p(X) = a_0 + a_1 X + \cdots + a_n X^n, \qquad a_j \in \mathbb{C}.$$

Two polynomials are equal when the coefficients of each power agree. Addition adds the coefficients of each power; multiplication by a complex number multiplies every coefficient. Polynomial multiplication expands brackets using $X^j X^k = X^{j+k}$ and collects equal powers. If $a_n \neq 0$, the degree is $\deg p = n$, and a_n is its leading coefficient. A monic polynomial has leading coefficient 1. The zero polynomial has every coefficient zero; we will not assign it a numerical degree here. The phrase *degree at most* n will also allow the zero polynomial.

NOTE

For nonzero polynomials p, q, we have $\deg(pq) = \deg p + \deg q$: the leading coefficient of the product is the product of the two nonzero leading coefficients. In particular, a product of nonzero polynomials is not the zero polynomial.

PROPOSITION A.1

For $p, d \in \mathbb{C}[X]$ with $d \neq 0$, there are unique polynomials q, r such that

$$p = dq + r, \qquad r = 0 \text{ or } \deg r < \deg d.$$

These are the quotient and remainder when dividing p by d. Long division repeatedly subtracts a multiple of d that cancels the leading term, so the degree decreases. For uniqueness, if $p = dq + r = dq' + r'$, then $d(q - q') = r' - r$. If $q - q' \neq 0$, the left side has degree at least $\deg d$, whereas the right side is zero or has smaller degree. Thus $q = q'$ and then $r = r'$.

PROPOSITION A.2

A number $a \in \mathbb{C}$ is a root of p when $p(a) = 0$. We have

$$p(a) = 0 \quad \Longleftrightarrow \quad p(X) = (X - a)q(X)$$

for some polynomial q.

Proof.
On division by $X - a$, the remainder is a constant r, so $p(X) = (X - a)q(X) + r$. Substitution at $X = a$ gives $r = p(a)$, proving both directions.

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PROPOSITION A.3

A nonzero polynomial of degree n has at most n distinct roots.

Proof.
For $n = 0$, the polynomial is a nonzero constant and has no roots. If there is a root a, write $p = (X - a)q$ with $\deg q = n - 1$. Every other root $b \neq a$ satisfies $0 = (b - a)q(b)$, so $q(b) = 0$. Induction gives at most $n - 1$ such roots, or at most n including a.

DEFINITION A.4

For $p \neq 0$ and an integer $m \geq 1$, a root a has multiplicity m if

$$p(X) = (X - a)^m q(X), \qquad q(a) \neq 0.$$

Thus exactly m factors $X - a$ can be removed. We do not assign a finite root multiplicity to the zero polynomial.

PROPOSITION A.4

The fundamental theorem of algebra. Every nonconstant complex polynomial has a complex root. Consequently, if $\deg p = n \geq 1$ and its leading coefficient is c, then

$$p(X) = c(X - \lambda_1) \cdots (X - \lambda_n).$$

In this list, each root occurs according to its multiplicity. We will use this theorem without proof.

NOTE

For $p(X) = \sum_{k=0}^{n} a_k X^k$, define $\overline{p}(X) = \sum_{k=0}^{n} \overline{a_k} X^k$ by conjugating the coefficients. Then $\overline{p}(\overline{z}) = \overline{p(z)}$. If p has real coefficients, then $\overline{p} = p$, so the conjugate $\overline{a}$ of every root a is also a root. Multiplicity is preserved as well: from $p = (X - a)^m q$ with $q(a) \neq 0$, coefficient conjugation gives

$$p(X) = (X - \overline{a})^m \overline{q}(X), \qquad \overline{q}(\overline{a}) = \overline{q(a)} \neq 0.$$

DEFINITION A.5

The formal derivative of $p(X) = \sum_{k=0}^{n} a_k X^k$ is

$$p'(X) = \sum_{k=1}^{n} k a_k X^{k-1}.$$

This is an operation on coefficients. The product rule $(fg)' = f'g + fg'$ holds: for $j, k \geq 1$, it is the identity $(X^j X^k)' = (j+k)X^{j+k-1} = jX^{j-1}X^k + X^j kX^{k-1}$; if one exponent is zero, the rule follows because a constant has derivative zero. Expansion and addition give the rule for all polynomials.

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PROPOSITION A.5

For $p \neq 0$, the root a has multiplicity at least 2 if and only if $p(a) = p'(a) = 0$.

Proof.
If $p(a) = 0$, write $p = (X - a)q$. The product rule gives

$$p' = q + (X - a)q', \qquad p'(a) = q(a).$$

Thus $p'(a) = 0$ is equivalent to q also being divisible by $X - a$, which means that p is divisible by $(X - a)^2$.

EXERCISE A.1

Please factor $P(X) = (X - 1)^4 + 4$ into linear factors over $\mathbb{C}$, and please find all its roots and their multiplicities.

WORKED SOLUTION

Set $Y = X - 1$. A difference of squares gives

$$Y^4 + 4 = (Y^2 + 2)^2 - (2Y)^2 = (Y^2 - 2Y + 2)(Y^2 + 2Y + 2).$$

Returning to X,

$$P(X) = (X^2 - 4X + 5)(X^2 + 1) = \big((X - 2)^2 + 1\big)(X^2 + 1).$$

Therefore

$$P(X) = (X - i)(X + i)(X - 2 - i)(X - 2 + i).$$

The roots are $i, -i, 2 + i, 2 - i$. They are distinct, so each has multiplicity 1.

EXERCISE A.2

Please find a real-coefficient polynomial of the smallest possible degree for which $1 + 2i$ is a root of multiplicity 2 and $p(0) = 50$. Please determine whether such a polynomial of smallest degree is unique.

WORKED SOLUTION

The conjugate $1 - 2i$ must also be a root of multiplicity 2. The two roots are distinct, so the degree is at least 4. In degree 4, the polynomial must be

$$p(X) = c(X - 1 - 2i)^2(X - 1 + 2i)^2 = c(X^2 - 2X + 5)^2.$$

The condition $p(0) = 50$ gives $25c = 50$, hence $c = 2$. Thus

$$p(X) = 2(X^2 - 2X + 5)^2 = 2X^4 - 8X^3 + 28X^2 - 40X + 50.$$

This polynomial satisfies all conditions. The root $1 + 2i$ has multiplicity exactly 2 because $1 + 2i \neq 1 - 2i$. The value at 0 determines c, so the polynomial of smallest degree is unique.

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EXERCISE A.3

Let $n \geq 0$ be an integer. Please find the remainder when X^{4n+3} is divided by $X^2 - 2X + 2$.

WORKED SOLUTION

The divisor has roots $\alpha = 1 + i$ and $\beta = 1 - i$. The remainder has the form $r(X) = aX + b$. Substituting the roots gives

$$a\alpha + b = \alpha^{4n+3}, \qquad a\beta + b = \beta^{4n+3}.$$

We compute

$$(1 \pm i)^2 = \pm 2i, \qquad (1 \pm i)^4 = -4, \qquad (1 \pm i)^3 = -2 \pm 2i.$$

Hence

$$a(1 + i) + b = (-4)^n(-2 + 2i), \qquad a(1 - i) + b = (-4)^n(-2 - 2i).$$

Subtracting gives $2ia = 4i(-4)^n$, so $a = 2(-4)^n$. Substitution in either equation gives $a + b = -2(-4)^n$, hence $b = -4(-4)^n$. The remainder is

$$r(X) = 2(-4)^n(X - 2).$$

It has degree less than 2 and the required values at both roots. Uniqueness of quotient and remainder gives the result.

EXERCISE A.4

Please find all $p \in \mathbb{C}[X]$ of degree at most 3 satisfying

$$p(i) = p'(i) = 0, \qquad p(-i) = 2, \qquad p'(-i) = 0.$$

WORKED SOLUTION

The conditions at i imply $p = (X - i)^2(aX + b)$ for some $a, b \in \mathbb{C}$. The value $p(-i) = 2$ gives

$$(-2i)^2(-ia + b) = 2 \quad \Longrightarrow \quad b - ia = -\frac{1}{2}.$$

Differentiation gives

$$p'(X) = 2(X - i)(aX + b) + a(X - i)^2.$$

Thus $p'(-i) = 0$ becomes

$$-4i(b - ia) - 4a = 0.$$

Substituting $b - ia = -1/2$ gives $2i - 4a = 0$, so $a = i/2$. Now $b - i(i/2) = -1/2$, hence $b = -1$. The unique polynomial is

$$p(X) = (X - i)^2 \left(\frac{i}{2}X - 1\right) = 1 + \frac{i}{2}(X^3 + 3X).$$

Indeed, $p(i) = 0$, $p(-i) = 2$, and $p'(X) = \frac{3i}{2}(X^2 + 1)$ vanishes at both points. Uniqueness also follows directly: the difference of two solutions would be divisible by $(X - i)^2(X + i)^2$, a degree-4 polynomial, although its degree is at most 3. The difference must therefore be zero.

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EXERCISE A.5

Let $p \in \mathbb{C}[X]$ have degree $n \geq 1$. Please prove that, except for at most $n-1$ values of $w \in \mathbb{C}$, the equation $p(z) = w$ has exactly n distinct solutions. Please find the exceptional values for $p(X) = X^3 - 3X$.

WORKED SOLUTION

The derivative p' has degree $n-1$, so it has at most $n-1$ distinct roots. Let

$$S = \{p(c) : c \in \mathbb{C},\ p'(c) = 0\}$$

be the set of possible exceptional values. Its number of elements is at most the number of roots of p', so $|S| \leq n-1$. If $w \notin S$, the polynomial $p - w$ has no repeated root: such a root c would satisfy both $p(c) = w$ and $p'(c) = 0$, contradicting the choice of w. The fundamental theorem of algebra gives n roots counted with multiplicity. They are all simple, hence n distinct solutions. Conversely, if $w \in S$, the polynomial $p-w$ has a repeated root and therefore fewer than n distinct roots. For the given example,

$$p'(X) = 3(X^2 - 1), \qquad p(1) = -2, \qquad p(-1) = 2.$$

The exceptional values are exactly $-2, 2$. Indeed,

$$p(X) - 2 = (X-2)(X+1)^2, \qquad p(X) + 2 = (X+2)(X-1)^2.$$

Each exceptional value has two distinct solutions; every other value has three.

EXERCISE A.6

Let $p(X) = X^n + a_{n-1}X^{n-1} + \cdots + a_0$ be monic, with $n \geq 1$, and set $a_n = 1$. Suppose all its roots lie on the unit circle. Please prove that

$$|a_0| = 1, \qquad a_k = a_0\overline{a_{n-k}} \quad (0 \leq k \leq n).$$

Please determine whether these coefficient conditions guarantee that all roots lie on the unit circle.

WORKED SOLUTION

Write $p(X) = (X-\lambda_1)\cdots(X-\lambda_n)$, listing each root according to its multiplicity, with $|\lambda_j| = 1$. The constant coefficient is

$$a_0 = (-1)^n \lambda_1 \cdots \lambda_n,$$

so $|a_0| = 1$. Define a new polynomial by reversing and conjugating the coefficients:

$$p^{\#}(X) = \sum_{k=0}^{n} \overline{a_{n-k}} X^k = (1 - \overline{\lambda_1}X)\cdots(1 - \overline{\lambda_n}X).$$

Since $\overline{\lambda_j}\lambda_j = 1$, each factor satisfies

$$1 - \overline{\lambda_j}X = -\overline{\lambda_j}(X - \lambda_j).$$

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EXERCISE A.6 CONTINUED

Consequently,

$$p^{\#}(X) = (-1)^n \overline{\lambda_1 \cdots \lambda_n}\, p(X) = \overline{a_0}\, p(X).$$

Comparison of coefficients of X^k gives $\overline{a_{n-k}} = \overline{a_0} a_k$. Multiplying by a_0 and using $a_0\overline{a_0} = 1$ gives the required identity. The conditions are not sufficient. For example,

$$p(X) = X^2 - \frac{5}{2}X + 1 = (X-2)\left(X - \frac{1}{2}\right)$$

satisfies them, but neither root lies on the unit circle.

EXERCISE A.7

Let V be a finite-dimensional complex inner product space, and let $T : V \to V$ be linear. Suppose every eigenvector of $T + T^*$ is also an eigenvector of $T - T^*$. Please prove that T is normal.

WORKED SOLUTION

Write

$$H := T + T^*, \qquad K := T - T^*.$$

We aim to find an orthonormal basis in which both maps act by scalar multiplication on every basis vector. First compute the adjoints:

$$H^* = (T + T^*)^* = T^* + (T^*)^* = T^* + T = H,$$

and

$$K^* = (T - T^*)^* = T^* - (T^*)^* = T^* - T = -K.$$

Thus H is self-adjoint. If $V = \{0\}$, the result is immediate. Otherwise, the spectral theorem provides an orthonormal basis $B = (e_1, \ldots, e_n)$ and real numbers $h_1, \ldots, h_n$ such that

$$He_j = h_j e_j \qquad (1 \le j \le n).$$

Every e_j is an eigenvector of H, so the assumption provides $k_j \in \mathbb{C}$ such that

$$Ke_j = k_j e_j.$$

The number k_j may depend on j; we need not assume that the eigenvalues of H are distinct. In the basis B, both matrices are diagonal:

$$[H]_B = D_H := \mathrm{diag}(h_1, \ldots, h_n), \qquad [K]_B = D_K := \mathrm{diag}(k_1, \ldots, k_n).$$

For every j,

$$HKe_j = H(k_j e_j) = k_j h_j e_j, \qquad KHe_j = K(h_j e_j) = h_j k_j e_j.$$

Scalars commute, so $HKe_j = KHe_j$ for every basis vector. Hence $HK = KH$ on all of V. Now recover T and T^*:

$$H + K = 2T, \qquad H - K = 2T^*, \qquad T = \frac{H+K}{2}, \quad T^* = \frac{H-K}{2}.$$

Compute the two products separately, without assuming in advance that the factors com-

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EXERCISE A.7 CONTINUED

mute:

$$\begin{aligned} TT^* &= \frac{1}{4}(H+K)(H-K) \\ &= \frac{1}{4}\big(HH - HK + KH - KK\big) \\ &= \frac{1}{4}\big(H^2 - HK + KH - K^2\big) \\ &= \frac{1}{4}(H^2 - K^2), \end{aligned}$$

where the last line uses $HK = KH$. Similarly,

$$\begin{aligned} T^*T &= \frac{1}{4}(H-K)(H+K) \\ &= \frac{1}{4}\big(HH + HK - KH - KK\big) \\ &= \frac{1}{4}\big(H^2 + HK - KH - K^2\big) \\ &= \frac{1}{4}(H^2 - K^2). \end{aligned}$$

Thus $TT^* = T^*T$, so T is normal.

We also make explicit the connection with unitary diagonalization and the order of the matrices. First choose a fixed orthonormal basis of V, write $A = [T]$ in that basis, and form U from the coordinate columns of $e_1, \dots, e_n$ in the same basis. Then U is unitary, and

$$U^*(A + A^*)U = D_H, \qquad U^*(A - A^*)U = D_K.$$

Adding the identities and dividing by 2 gives

$$U^*AU = D := \frac{1}{2}(D_H + D_K), \qquad A = UDU^*.$$

Thus $[T]_B = D$ and $[T^*]_B = D^*$. For diagonal matrices,

$$(DD^*)_{ij} = \sum_{\ell=1}^{n} d_{i\ell}\overline{d_{j\ell}} = \begin{cases} |d_{ii}|^2, & i = j, \\ 0, & i \neq j, \end{cases}$$

and

$$(D^*D)_{ij} = \sum_{\ell=1}^{n} \overline{d_{\ell i}}\, d_{\ell j} = \begin{cases} |d_{ii}|^2, & i = j, \\ 0, & i \neq j. \end{cases}$$

We can therefore check the equality again in matrix form:

$$\begin{aligned} AA^* &= (UDU^*)(UD^*U^*) = UD(U^*U)D^*U^* = U(DD^*)U^*, \\ A^*A &= (UD^*U^*)(UDU^*) = UD^*(U^*U)DU^* = U(D^*D)U^*. \end{aligned}$$

The two results agree. The basis that diagonalizes the self-adjoint part also diagonalizes the other part by the hypothesis, and consequently diagonalizes T itself.

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EXERCISE A.8

Let $\mathbb{F} \in \{\mathbb{R}, \mathbb{C}\}$ and $A \in M_n(\mathbb{F})$. For any matrix C, please write $C^* := \overline{C}^T$; over $\mathbb{R}$, $C^* = C^T$. Following the convention that the inner product is linear in its first argument, define

$$\langle x, y \rangle := y^* x = x^T \overline{y}.$$

1. Please prove that $\langle Av, w \rangle = \langle v, A^* w \rangle$ for all $v, w \in \mathbb{F}^n$.
2. Let $G \in M_n(\mathbb{F})$ be Hermitian and positive definite. Define

   $$\langle x, y \rangle_G := y^* G x.$$

   Please prove that this is an inner product, and please show that there is a unique matrix A^{*_G} satisfying

   $$\langle Av, w \rangle_G = \langle v, A^{*_G} w \rangle_G \quad \text{for all } v, w \in \mathbb{F}^n,$$

   and that $A^{*_G} = G^{-1} A^* G$.

WORKED SOLUTION

1. The identity $(A^* w)^* = w^* A$ gives

   $$\langle Av, w \rangle = w^* A v = (A^* w)^* v = \langle v, A^* w \rangle.$$

2. Linearity in the first argument follows from the linearity of matrix multiplication. Since $G^* = G$,

   $$\overline{\langle y, x \rangle_G} = \overline{x^* G y} = (x^* G y)^* = y^* G^* x = y^* G x = \langle x, y \rangle_G.$$

   Furthermore, $\langle x, x \rangle_G = x^* G x > 0$ for every $x \neq 0$, and $\langle 0, 0 \rangle_G = 0$. The matrix G is invertible: otherwise $Gx = 0$ for some $x \neq 0$ would imply $x^* G x = 0$. Set $B := G^{-1} A^* G$. Then

   $$B^* = (G^{-1} A^* G)^* = G A G^{-1},$$

   and hence

   $$\langle v, Bw \rangle_G = (Bw)^* G v = w^* B^* G v = w^* G A v = \langle Av, w \rangle_G.$$

   For uniqueness, if C also satisfies the identity, then $w^*(C^* G - G A)v = 0$ for all v, w. Choosing $v = e_j$ and $w = e_i$ shows that every entry of $C^* G - G A$ is zero. Thus

   $$C^* G = G A \implies C^* = G A G^{-1} \implies C = G^{-1} A^* G = B.$$

A.3 Unitary Diagonalization

NOTATION AND CONVENTIONS

The following exercises are over $\mathbb{C}$, and $A^* = \overline{A}^T$ is the conjugate transpose. A matrix U is unitary when $U^* U = U U^* = I$. A unitary diagonalization has the form $U^* A U = D$ with D diagonal. The inner product is linear in its first variable.

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PROPOSITION A.6

A matrix $A \in M_n(\mathbb{C})$ is unitarily diagonalizable if and only if it is normal, meaning $AA^* = A^*A$.

Proof.
If $A = UDU^*$, then $A^* = UD^*U^*$, so

$$AA^* = UDD^*U^*, \qquad A^*A = UD^*DU^*.$$

Diagonal matrices commute, giving $AA^* = A^*A$. Conversely, suppose A is normal, and set

$$H = \frac{A + A^*}{2}, \qquad K = \frac{A - A^*}{2i}.$$

We have $H^* = H$ and

$$K^* = \frac{-1}{2i}(A^* - A) = K,$$

so both are Hermitian. Compute both products:

$$HK = \frac{1}{4i}\big(A^2 - AA^* + A^*A - (A^*)^2\big),$$
$$KH = \frac{1}{4i}\big(A^2 + AA^* - A^*A - (A^*)^2\big).$$

Normality gives $HK = KH$. By the Hermitian spectral theorem, the space is the orthogonal direct sum of the eigenspaces of H. If $Hx = \lambda x$, then

$$H(Kx) = K(Hx) = \lambda Kx,$$

so K preserves each eigenspace of H. Its restriction to each such space is Hermitian, since $\langle Kx, y\rangle = \langle x, Ky\rangle$ there. Choose an orthonormal eigenbasis of K in each space. Their union is a common orthonormal eigenbasis of H, K, and hence of $A = H + iK$. Its basis matrix U is unitary and satisfies $U^*AU = D$ diagonal. For $n = 0$, the empty basis proves the statement immediately.

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EXERCISE A.9 CHALLENGE

Consider

$$A = \begin{pmatrix} 1 & 2 \\ 0 & -1 \end{pmatrix}.$$

Please compute A^2. Please find a diagonalization $P^{-1}AP = D$, and please determine whether a unitary diagonalization also exists. Please determine whether $A^2 = I$ together with all eigenvalues on the unit circle forces A to be unitary.

WORKED SOLUTION

Direct multiplication gives

$$A^2 = \begin{pmatrix} 1 & 2-2 \\ 0 & 1 \end{pmatrix} = I_2.$$

Eigenvectors for $1, -1$ are $(1,0)^T$ and $(-1,1)^T$, respectively. Thus take

$$P = \begin{pmatrix} 1 & -1 \\ 0 & 1 \end{pmatrix}, \quad P^{-1} = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}.$$

Then

$$AP = \begin{pmatrix} 1 & 1 \\ 0 & -1 \end{pmatrix}, \qquad P^{-1}AP = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

However,

$$A^*A = \begin{pmatrix} 1 & 2 \\ 2 & 5 \end{pmatrix}, \qquad AA^* = \begin{pmatrix} 5 & -2 \\ -2 & 1 \end{pmatrix}.$$

They differ, so A is not normal and has no unitary diagonalization. In particular, $A^*A \neq I_2$, so A is not unitary despite $A^2 = I_2$ and its eigenvalues $1, -1$ lying on the unit circle. Its eigendirections are not orthogonal: $\langle (1,0), (-1,1) \rangle = -1$.

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EXERCISE A.10 CHALLENGE

Let $U \in M_n(\mathbb{C})$ be unitary, with $n \geq 1$.

1. Please prove that there is a unitary matrix S satisfying $S^2 = U$.
2. If U has n distinct eigenvalues, please find the number of unitary matrices S satisfying $S^2 = U$.
3. Please construct infinitely many distinct unitary matrices satisfying $S^2 = -I_2$.

WORKED SOLUTION

1. A unitary matrix is normal, so $U = V\,\mathrm{diag}(\lambda_1, \ldots, \lambda_n)V^*$ with V unitary. If $Uq_j = \lambda_j q_j$ and $q_j \neq 0$, then
$$\|q_j\| = \|Uq_j\| = |\lambda_j|\|q_j\|,$$
so $|\lambda_j| = 1$. Write $\lambda_j = e^{i\theta_j}$, and set
$$S = V\,\mathrm{diag}(e^{i\theta_1/2}, \ldots, e^{i\theta_n/2})V^*.$$
Every diagonal entry has modulus 1, giving $S^*S = SS^* = I$ and $S^2 = U$.

2. Every square root commutes with U, since $SU = S^3 = US$. Thus S preserves every eigenspace of U. Distinct eigenvalues make these spaces one-dimensional, so in the same eigenbasis
$$S = V\,\mathrm{diag}(\mu_1, \ldots, \mu_n)V^*, \qquad \mu_j^2 = \lambda_j.$$
Each nonzero λ_j has exactly two distinct complex square roots, both of modulus 1. The choices are independent, giving exactly 2^n such matrices.

3. For $0 \leq t < \pi$, let $v_t = (\cos t, \sin t)^T$ and let $P_t = v_t v_t^T$ be the orthogonal projection onto its span. We have $P_t^2 = P_t = P_t^*$. Set
$$S_t = i(2P_t - I_2) = i\begin{pmatrix} \cos(2t) & \sin(2t) \\ \sin(2t) & -\cos(2t) \end{pmatrix}.$$
Since $(2P_t - I_2)^2 = 4P_t^2 - 4P_t + I_2 = I_2$,
$$S_t^2 = -I_2, \qquad S_t^* S_t = (-i)(i)(2P_t - I_2)^2 = I_2.$$
The lines $\mathrm{span}\{v_t\}$ are distinct for $t \in [0, \pi)$, so their projections P_t and the matrices S_t are distinct. This gives an infinite family.

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EXERCISE A.11 CHALLENGE

Let $A, B \in M_n(\mathbb{C})$ be normal.

1. Please prove that there is a unitary Q for which both Q^*AQ and Q^*BQ are diagonal if and only if $AB = BA$.

2. Please find such a simultaneous diagonalization for

$$A = \mathrm{diag}(1+i, 1+i, -2i), \qquad B = \begin{pmatrix} 2 & i & 0 \\ -i & 2 & 0 \\ 0 & 0 & 1+i \end{pmatrix}.$$

   Please also compute Q^*ABQ.

WORKED SOLUTION

1. If $A = QD_AQ^*$ and $B = QD_BQ^*$, then

$$AB = QD_AD_BQ^* = QD_BD_AQ^* = BA.$$

   Conversely, suppose $AB = BA$. Unitarily diagonalize $A = VDV^*$, and set $C = V^*BV$. Entrywise, $DC = CD$ gives

$$(\lambda_i - \lambda_j)c_{ij} = 0.$$

   Thus C is block diagonal, with one block per distinct eigenvalue of A. Normality of B gives $CC^* = C^*C$, so every block is normal. Unitarily diagonalize each block, and combine the changes into a block-diagonal unitary matrix W. Then W^*CW is diagonal, while $W^*DW = D$, since each block of D is scalar. The matrix $Q = VW$ gives both diagonalizations.

2. The diagonal matrix A is normal. Both blocks of B are normal: the first is Hermitian and the second is scalar. The matrices commute because the first block of A is $(1+i)I_2$. Choose

$$q_1 = \frac{1}{\sqrt{2}}(i, 1, 0)^T, \quad q_2 = \frac{1}{\sqrt{2}}(-i, 1, 0)^T, \quad q_3 = (0, 0, 1)^T.$$

   Their norms are 1, and $\langle q_1, q_2 \rangle = (i \cdot i + 1)/2 = 0$; also q_3 is orthogonal to both. Direct multiplication gives

$$Bq_1 = 3q_1, \qquad Bq_2 = q_2, \qquad Bq_3 = (1+i)q_3.$$

   For $Q = (q_1\ q_2\ q_3)$, we obtain

$$\begin{aligned} Q^*AQ &= \mathrm{diag}(1+i, 1+i, -2i), \\ Q^*BQ &= \mathrm{diag}(3, 1, 1+i), \\ Q^*ABQ &= \mathrm{diag}(3+3i, 1+i, 2-2i). \end{aligned}$$

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EXERCISE A.12 CHALLENGE

Let $A \in M_n(\mathbb{C})$ be normal, and suppose $\|A^2x\| = \|Ax\|$ for every $x \in \mathbb{C}^n$. Please prove that A^*A is an orthogonal projection matrix and that $\operatorname{Im} A = (\ker A)^\perp$. Please prove that the restriction of A to $\operatorname{Im} A$ is unitary. Please check whether normality can be dropped from the conclusion about A^*A.

WORKED SOLUTION

Write $A = UDU^*$ in an orthonormal eigenbasis. For a unit eigenvector q_j with eigenvalue λ_j, the assumption gives

$$|\lambda_j|^2 = \|A^2 q_j\| = \|Aq_j\| = |\lambda_j|.$$

Thus $|\lambda_j|$ is 0 or 1. With $\varepsilon_j = |\lambda_j|^2$,

$$A^*A = U \operatorname{diag}(\varepsilon_1, \ldots, \varepsilon_n) U^*, \qquad \varepsilon_j \in \{0, 1\}.$$

This matrix is Hermitian and satisfies $(A^*A)^2 = A^*A$, so it is an orthogonal projection. The kernel of A is spanned by the eigenvectors with $\lambda_j = 0$, and its image by the remaining eigenvectors: for $\lambda_j \neq 0$,

$$q_j = A(\lambda_j^{-1} q_j).$$

Orthonormality gives $\operatorname{Im} A = (\ker A)^\perp$. On the image, the restriction is diagonal with all entries of modulus 1, hence unitary. If the image is the zero space, the empty basis gives the assertion. Without normality, take

$$C = \begin{pmatrix} 1 & 3 \\ 0 & 0 \end{pmatrix}.$$

Then $C^2 = C$, so $\|C^2x\| = \|Cx\|$ for every x. However,

$$C^*C = \begin{pmatrix} 1 & 3 \\ 3 & 9 \end{pmatrix}, \qquad (C^*C)^2 = 10C^*C \neq C^*C.$$

Thus C^*C is not a projection.

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EXERCISE A.13 CHALLENGE

For a matrix $M = (m_{jk}) \in M_n(\mathbb{C})$, define its Frobenius norm by

$$\|M\|_F^2 = \sum_{j,k=1}^{n} |m_{jk}|^2 = \mathrm{tr}(M^* M).$$

Let $A \in M_n(\mathbb{C})$ be normal and invertible.

1. Please find the unitary matrix U minimizing $\|A - U\|_F$, and please prove that it is unique.
2. Please compute it and the minimum distance for

$$A = \frac{1}{2}\begin{pmatrix} -3+2i & 3+2i \\ 3+2i & -3+2i \end{pmatrix}.$$

Hint: please use an orthonormal eigenbasis of A, and please use $|w_{jj}| \leq 1$ for every unitary matrix W.

WORKED SOLUTION

The Frobenius norm is unchanged by multiplication by unitary matrices. For example, if V is unitary,

$$\|V^* M V\|_F^2 = \mathrm{tr}(V^* M^* V V^* M V) = \mathrm{tr}(V^* M^* M V) = \mathrm{tr}(M^* M).$$

The last step uses $\mathrm{tr}(XY) = \mathrm{tr}(YX)$, obtained by summing the entries $\sum_{j,k} x_{jk} y_{kj}$ in either order. Write $A = VDV^*$ with $D = \mathrm{diag}(\lambda_1, \ldots, \lambda_n)$ and every $\lambda_j \neq 0$. For each unitary U, set $W = V^* U V$, which is also unitary. Then

$$\begin{aligned} \|A - U\|_F^2 &= \|D - W\|_F^2 \\ &= \sum_j |\lambda_j|^2 + n - 2\,\mathrm{Re} \sum_j \lambda_j \overline{w_{jj}} \\ &\geq \sum_j |\lambda_j|^2 + n - 2 \sum_j |\lambda_j| \\ &= \sum_j (|\lambda_j| - 1)^2. \end{aligned}$$

Here $\sum_{j,k} |w_{jk}|^2 = n$ because each column of W is a unit vector; in particular $|w_{jj}| \leq 1$. Equality is attained by

$$W_0 = \mathrm{diag}\left(\frac{\lambda_1}{|\lambda_1|}, \ldots, \frac{\lambda_n}{|\lambda_n|}\right), \qquad U_0 = V W_0 V^*.$$

For uniqueness, every bound $\mathrm{Re}(\lambda_j \overline{w_{jj}}) \leq |\lambda_j|$ must be an equality. As $\lambda_j \neq 0$, this forces $w_{jj} = \lambda_j / |\lambda_j|$, hence $|w_{jj}| = 1$. The squared moduli in that row sum to 1, so every other entry in the row is zero. Thus $W = W_0$ and $U = U_0$. In the numerical case, take

$$V = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}, \qquad V^* A V = \mathrm{diag}(2i, -3).$$

Therefore

$$U_0 = V\,\mathrm{diag}(i, -1) V^* = \frac{1}{2}\begin{pmatrix} -1+i & 1+i \\ 1+i & -1+i \end{pmatrix},$$

and the distance is $\sqrt{(2-1)^2 + (3-1)^2} = \sqrt{5}$.

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A.4 Matrices and Linear Maps: Reference Table

NOTATION AND CONVENTIONS

In this table, V, W are finite-dimensional spaces over $\mathbb{F}$, with dimensions n, m and ordered bases B, C. For $T : V \to W$, write $A = [T]_C^B \in M_{m\times n}(\mathbb{F})$. Rows involving a subspace U assume $U \le V$, and $[U]_B = \{[u]_B : u \in U\}$ denotes its coordinate set. Square-matrix properties refer to $T : V \to V$, using the same basis in domain and codomain. Inner-product rows are over $\mathbb{R}$ or $\mathbb{C}$; identities involving A^* use orthonormal bases, except for the row explicitly introducing the matrix G. The symbol I denotes an identity, and 0 denotes a zero vector, map or matrix as appropriate. The symbols ker, Im, Row and Col denote kernel, image, row space and column space. For the zero space the basis is empty, rank and nullity are 0, and the empty matrix has determinant 1 and trace 0.

Concept	Maps and spaces	Matrix form
Vector and coordinates	$v = \sum_j x_j b_j$; the coordinates are its coefficients in the basis B.	$[v]_B = (x_1, \dots, x_n)^T$.
Linear combination	A vector $\sum_j \alpha_j v_j$ with scalars $\alpha_j \in \mathbb{F}$.	If the columns of M are $[v_j]_B$, then $[\sum_j \alpha_j v_j]_B = M\alpha$.
Span	$\operatorname{span}\{v_j\}$ is the set of all linear combinations of the list.	The column space of its coordinate matrix.
Linear independence	$\sum_j \alpha_j v_j = 0$ forces every $\alpha_j = 0$.	The columns of M are independent exactly when $\ker M = \{0\}$.
Basis	An independent list spanning the space.	The coordinate matrix of a basis of $\mathbb{F}^n$ is square and invertible.
Dimension	$\dim V$ is the number of vectors in any basis of V.	$\dim \mathbb{F}^n = n$; $\dim M_{m\times n}(\mathbb{F}) = mn$.
Linear map	$T(\alpha v + \beta w) = \alpha T(v) + \beta T(w)$.	$[T(v)]_C = A[v]_B$.
Representing matrix	Column j gives the coordinates of $T(b_j)$ in C.	$A = [T]_C^B$. $A_{\cdot j} = [T(b_j)]_C$.
Composition	$S \circ T$ applies T first and then S.	$[S \circ T]_D^B = [S]_D^C [T]_C^B$; the intermediate bases must agree.
Kernel	$\ker T = \{v \in V : T(v) = 0\}$.	$[\ker T]_B = \ker A$; the solutions of $Ax = 0$.
Image	$\operatorname{Im} T = \{T(v) : v \in V\}$.	$[\operatorname{Im} T]_C = \operatorname{Col} A$.
Rank	$\operatorname{rank} T = \dim \operatorname{Im} T$.	$\operatorname{rank} A = \dim \operatorname{Col} A = \dim \operatorname{Row} A$.
Nullity	$\operatorname{nullity} T = \dim \ker T$.	$\operatorname{nullity} A = n - \operatorname{rank} A$.
Rank-nullity	$\dim \ker T + \dim \operatorname{Im} T = n$.	The number of free variables plus the rank equals the number of columns.
Injectivity	$T(v) = T(w)$ implies $v = w$; equivalently $\ker T = \{0\}$.	The columns of A are independent; $\operatorname{rank} A = n$.
Surjectivity	$\operatorname{Im} T = W$.	The columns of A span $\mathbb{F}^m$; $\operatorname{rank} A = m$.
Isomorphism	A linear map that is both injective and surjective.	$m = n$ and A is invertible.
Inverse	$T^{-1}T = I_V$ and $TT^{-1} = I_W$.	$[T^{-1}]_B^C = A^{-1}$; $AA^{-1} = A^{-1}A = I$. A square matrix is singular when it is not invertible, equivalently $\det A = 0$.

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Concept	Maps and spaces	Matrix form
Row space	The subspace spanned by the rows, viewed as vectors in $\mathbb{F}^n$.	Row $A \leq \mathbb{F}^n$; row operations preserve it.
Column space	The subspace spanned by the columns, in $\mathbb{F}^m$.	Col $A = \mathrm{Im}(x \mapsto Ax)$.
Row equivalence	A sequence of invertible row operations; applied to an augmented system, they preserve its solutions.	$B = EA$ with E invertible, a product of elementary matrices.
Change of basis	P converts B' coordinates into B coordinates.	$[v]_B = P[v]_{B'}$; column j is $[b'_j]_B$.
Similarity	Two matrices of the same endomorphism $T : V \to V$ in different bases.	$D = P^{-1}AP$ with P invertible.
Trace	A change of basis preserves the sum of diagonal entries.	$\mathrm{tr}\, A = \sum_j a_{jj}$; $\mathrm{tr}(XY) = \mathrm{tr}(YX)$ when both products are defined and square.
Determinant	A scalar attached to an endomorphism, independent of the chosen basis.	$\det A$; $\det(AB) = \det A \det B$. Over $\mathbb{R}$, its absolute value is the volume scale factor.
Minor and cofactor	For $n \geq 1$, $A_{\widehat{i},\widehat{j}}$ is obtained by deleting row i and column j.	$C_{ij} = (-1)^{i+j} \det A_{\widehat{i},\widehat{j}}$.
Classical adjugate	The transpose of the cofactor matrix; here $n \geq 1$.	$\mathrm{adj}(A) = (C_{ij})^T$; $A\,\mathrm{adj}(A) = \det(A)I$.
Eigenvalue and eigenvector	$T(v) = \lambda v$ with $v \neq 0$.	$Ax = \lambda x$ with $x \neq 0$; $\det(A - \lambda I) = 0$.
Eigenspace	$E_\lambda(T) = \ker(T - \lambda I_V)$.	$E_\lambda(A) = \ker(A - \lambda I_n)$.
Characteristic polynomial	Independent of basis; its roots in the field are the eigenvalues.	$p_A(t) = \det(tI_n - A)$; a monic polynomial of degree n.
Algebraic multiplicity	The multiplicity of λ as a root of the characteristic polynomial.	If $p_A(t) = (t - \lambda)^r q(t)$ and $q(\lambda) \neq 0$, the multiplicity is r.
Geometric multiplicity	The dimension of $E_\lambda(T)$.	$\dim \ker(A - \lambda I)$; between 1 and the algebraic multiplicity for an eigenvalue λ.
Diagonalization	Existence of a basis of eigenvectors of T.	$P^{-1}AP = D$ diagonal. Equivalent to splitting of the characteristic polynomial over the field and equality of both multiplicities at every eigenvalue.
Cayley-Hamilton	Substituting the endomorphism into its characteristic polynomial gives the zero map.	$p_A(A) = 0$; a constant term c becomes cI_n.
Minimal polynomial	The monic polynomial of least degree annihilating the endomorphism.	$m_A(A) = 0$ and $m_A \mid p_A$; for $n \geq 1$.
Nilpotence	$T^k = 0$ for some $k \geq 1$.	$A^k = 0$ for some $k \geq 1$; every eigenvalue is zero.

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Concept	Maps and spaces	Matrix form
Idempotence	$T^2 = T$; the map acts as the identity on its image.	$A^2 = A$; projection onto the image along the kernel.
Involution	$T^2 = I_V$; applying the map twice recovers the input.	$A^2 = I_n$; $A^{-1} = A$.
Zero and identity	The zero map sends every vector to zero; I_V fixes every vector.	$0_{m\times n}$ has only zero entries. The diagonal of I_n consists of ones, and all other entries are zero.
Diagonal matrix	In the basis B, $T(b_j) = d_j b_j$ for every j.	$D = \operatorname{diag}(d_1, \ldots, d_n)$; every off-diagonal entry is zero.
Triangular matrix	For $1 \le k \le n$, upper triangular preserves every $\operatorname{span}\{b_1, \ldots, b_k\}$; lower triangular preserves every $\operatorname{span}\{b_k, \ldots, b_n\}$.	Upper: $a_{ij} = 0$ for $i > j$. Lower: $a_{ij} = 0$ for $i < j$.
Scalar matrix	$T(v) = \lambda v$ for every $v \in V$, for a fixed $\lambda \in \mathbb{F}$.	$A = \lambda I_n$. Its matrix is the same in every basis of V.
Invariant subspace	$U \le V$ satisfies $T(U) \subseteq U$. A basis of U can be extended to a basis of V.	In such a basis, partitioned as $k + (n-k)$ with $k = \dim U$: $\begin{pmatrix} A_{11} & A_{12} \\ 0 & A_{22} \end{pmatrix}$.
Elementary matrix	Represents one elementary row operation. A row-scaling factor must be nonzero; every such operation is invertible.	E is obtained from I_m by one operation. The product EA applies that operation to the rows of A.
Permutation matrix	Permutes the basis vectors: $T(e_j) = e_{\sigma(j)}$, where σ permutes $\{1, \ldots, n\}$.	Every row and column has one entry equal to 1 and all others zero. $P^{-1} = P^T$.
Transpose	Interchange the row and column indices.	$(A^T)_{ij} = a_{ji}$; $(AB)^T = B^T A^T$.
Conjugate transpose	Over $\mathbb{R}$ or $\mathbb{C}$, transpose and conjugate every entry.	$(A^*)_{ij} = \overline{a_{ji}}$; $(AB)^* = B^* A^*$.
Adjoint map	$\langle Tv, w\rangle_W = \langle v, T^* w\rangle_V$ for all v, w.	In orthonormal bases: $[T^*]_B^C = A^*$.
Inner product	Linear in its first variable, conjugate symmetric and positive definite.	In an orthonormal basis: $\langle v, w\rangle = [w]^*[v]$.
Inner-product matrix	In a basis $B = (b_j)$, set $G_{ij} = \langle b_j, b_i\rangle$. Then $\langle v, w\rangle = [w]_B^* G [v]_B$.	In the same basis, for $T : V \to V$: $[T^*]_B = G^{-1} A^* G$.
Induced norm	$\|v\| = \sqrt{\langle v, v\rangle}$.	In an orthonormal basis: $\|v\|^2 = \sum_j |x_j|^2$.
Orthogonality	$v \perp w$ means $\langle v, w\rangle = 0$.	In an orthonormal basis: $[w]^*[v] = 0$.
Orthonormal basis	$\langle e_i, e_j\rangle = \delta_{ij}$; $\delta_{ij} = 1$ if $i = j$, and zero otherwise.	The basis matrix Q satisfies $Q^* Q = I$.
Orthogonal complement	$U^\perp = \{v : \langle v, u\rangle = 0 \text{ for all } u \in U\}$.	If the columns of M span U in orthonormal coordinates, then $[U^\perp]_B = \ker M^*$.

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Concept	Maps and spaces	Matrix form
Orthogonal projection	$P_U v \in U$ and $v - P_U v \in U^\perp$.	$P^2 = P = P^*$; if the columns of Q are an orthonormal basis of U, then $P = QQ^*$.
Gram-Schmidt	Subtract projections onto preceding directions and normalize each nonzero residual.	For independent input: $w_k = v_k - \sum_{j<k} \langle v_k, e_j \rangle e_j$; $e_k = w_k / \|w_k\|$.
Linear isometry	$\langle Tv, Tw \rangle = \langle v, w \rangle$ for all v, w.	In orthonormal bases: $A^*A = I_n$, even when A is rectangular.
Unitarity	An invertible inner-product-preserving endomorphism over $\mathbb{C}$.	$A^*A = AA^* = I_n$; $A^{-1} = A^*$.
Orthogonal matrix	An invertible isometry of a real inner product space.	$A^TA = AA^T = I_n$; $A^{-1} = A^T$.
Self-adjoint and Hermitian	$T = T^*$; every eigenvalue is real.	$A = A^*$ in an orthonormal basis. Over $\mathbb{R}$: $A = A^T$.
Symmetry	The algebraic matrix condition $A^T = A$. Over $\mathbb{R}$, it represents self-adjointness in an orthonormal basis.	Over $\mathbb{C}$, $A^T = A$ differs from $A^* = A$.
Skew-adjointness	$T^* = -T$; over $\mathbb{C}$ every eigenvalue is purely imaginary.	$A^* = -A$ in an orthonormal basis.
Skew-symmetry	The matrix condition $A^T = -A$; in characteristic different from 2, the diagonal is zero.	Over $\mathbb{R}$, in an orthonormal basis, it represents skew-adjointness.
Normality	$TT^* = T^*T$.	$AA^* = A^*A$ in an orthonormal basis; over $\mathbb{C}$, equivalent to unitary diagonalizability.
Positive definiteness	$T = T^*$ and $\langle Tv, v \rangle > 0$ for all $v \neq 0$.	$A = A^*$ and $x^*Ax > 0$ for all $x \neq 0$; every eigenvalue is positive.
Positive semidefiniteness	$T = T^*$ and $\langle Tv, v \rangle \geq 0$ for all v.	$A = A^*$ and $x^*Ax \geq 0$ for all x; every eigenvalue is nonnegative.
Unitary diagonalization	Existence of an orthonormal eigenbasis in a complex inner product space.	$U^*AU = D$ diagonal with U unitary; equivalent to normality.
Orthogonal diagonalization	Existence of a real orthonormal eigenbasis.	$Q^TAQ = D$ real diagonal with Q orthogonal; equivalent to symmetry of $A \in M_n(\mathbb{R})$.
Frobenius norm	A norm on matrix space: the square root of the sum of squared moduli of all entries.	$\|A\|_F^2 = \sum_{i,j} \|a_{ij}\|^2 = \mathrm{tr}(A^*A)$.

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APPENDIX B

Logic and Proof Methods

B.1 Logic

This appendix introduces some foundations of mathematical logic. We will study propositions, quantifiers, and logical connectives, and see how they are used to construct proofs and assess mathematical claims.

B.1.1 Propositions

A proposition is a statement to which exactly one of the truth values ``true'' and ``false'' can be assigned unambiguously. Examples of propositions include:

- Rome is the capital of Italy. (A true proposition.)
- $3^2 = 8$. (A false proposition.)

Consider the statement that the number of grains of sand on Earth exceeds 10^{24}. We may not know its truth value, and checking it would be difficult. Nevertheless, under an agreed meaning of the statement, it is either true or false; our lack of knowledge does not itself prevent it from being a proposition.

Examples of expressions that are not propositions include:

- How are you? (A question is not a proposition.)
- Please close the door! (An instruction is not a proposition.)
- $x^3 > x$. (No truth value is determined until the meaning or quantification of x has been specified.)

The sentence ``The Mona Lisa is a beautiful painting'' expresses a subjective judgement. To treat it as a formal proposition, one would first need to specify a precise meaning of ``beautiful.'' In mathematical logic, we work with statements whose meanings and domains of discourse have been specified.

B.1.2 Quantifiers

- Consider ``He is the prime minister.'' Without specifying the referent, this does not determine a proposition. Similarly, $n > 4$ does not determine a truth value when n is unspecified. To bind variables in statements, we use two logical symbols called **quantifiers**.

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RECALL AND TOOLS

- **For every**, denoted by $\forall$. For example, ``Every natural number is greater than 4'' is written as
$$\forall n \in \mathbb{N}\ (n > 4).$$
- **There exists**, denoted by $\exists$. For example, ``There exists a natural number greater than 4'' is written as
$$\exists n \in \mathbb{N}\ (n > 4).$$

Specify the domain of each quantifier, either explicitly or through a clearly established context. The following example explains why this matters.

- The expression $\forall x\ x^3 \geq x$ is incomplete without an agreed domain. The operations and comparison may not even be defined for the objects under consideration. In contrast, the following are propositions because the domain is stated:
  - $\forall x \in \mathbb{N}\ x^3 \geq x$ is true, since $x^3 \geq x$ for every natural number x.
  - $\forall x \in \mathbb{Q}\ x^3 \geq x$ is false. Choosing $x = \frac{1}{3}$ gives $x^3 = \frac{1}{27} < \frac{1}{3} = x$, a counterexample.

EXERCISE **B.1**

Please write the following statement in mathematical notation: a set is contained in another set if and only if every element of the first set also belongs to the second set.

WORKED SOLUTION

$$A \subseteq B \iff \forall a \in A,\ a \in B.$$

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Logical Connectives: Conjunction, Disjunction, and Negation

Conjunction ($\wedge$): both conditions hold.

- Saying that ``our room is large and well lit'' means that ``our room is large'' *and* ``our room is well lit.'' The conjunction is true exactly when both statements are true. If either statement is false, the conjunction is false.

Truth table: The table gives the truth value of $A \wedge B$ for every possible pair of truth values of A, B.

A	B	$A \wedge B$
T	T	T
T	F	F
F	T	F
F	F	F

Disjunction ($\vee$): logical ``or.''

- $A \vee B$ is true when at least one of A, B is true, and false when both are false. This is **inclusive** ``or'': it permits both statements to be true.

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Truth table:

A	B	$A \vee B$
T	T	T
T	F	T
F	T	T
F	F	F

Negation ($\neg$): reversing a truth value.

- For example, the negation of ``Every multiple of 6 is a multiple of 3'' is ``Not every multiple of 6 is a multiple of 3,'' or equivalently, ``There exists a multiple of 6 that is not a multiple of 3.'' A proposition and its negation have opposite truth values: one is true exactly when the other is false.

Truth table:

A	$\neg A$
T	F
F	T

EXERCISE B.2

Please use logical quantifiers to write the following proposition and its negation: there exists a natural number M such that $n \leq M$ for every natural number n.

WORKED SOLUTION

The proposition is

$$\exists M \in \mathbb{N}\, \forall n \in \mathbb{N},\ n \leq M.$$

Its negation is

$$\forall M \in \mathbb{N}\, \exists n \in \mathbb{N},\ n > M.$$

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Implication ($\Rightarrow$) and Equivalence ($\Leftrightarrow$)

**Implication ($\Rightarrow$): ``if … then ….''** Consider the statement ``If Tamar received a final grade of at least 75, then Tamar passed the course.'' Let A denote the assumption and B the conclusion.

- If A is true and B is true, the implication is true.
- If A is true and B is false, the implication is false.
- If A is false and B is true, the implication is true.
- If A is false and B is false, the implication is true.

This example illustrates the meaning of implication.

The implication is **false** only when A, the assumption, is true and B, the conclusion, is false. In every other case it is true.

Truth table:

A	B	$A \Rightarrow B$
T	T	T
T	F	F
F	T	T
F	F	T

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Example: The proposition

$$4 = 7 \Rightarrow 6 = 9$$

is **true**: its assumption is false, even though its conclusion is also false.

Important remarks. For a statement of the form ``If $P$, then $Q$,'' we ask whether $Q$ must hold whenever $P$ holds. Everyday language often uses ``if … then …'' causally (for example, ``If you eat, you will feel full''). Formal implication has the truth-table meaning given above; it need not express a causal connection.

Negating an implication. The implication fails exactly when P is true and Q is false:

$$\neg(P \Rightarrow Q) \equiv P \wedge \neg Q.$$

In words: The statement $P \Rightarrow Q$ says that whenever P is true, Q is true as well. To negate it, identify the case in which it fails: P is true but Q is false.

Equivalence ($\Leftrightarrow$): ``if and only if.''

- $P \Leftrightarrow Q$ is true exactly when P, Q have the same truth value: either both are true or both are false.

Truth table:

A	B	$A \Leftrightarrow B$
T	T	T
T	F	F
F	T	F
F	F	T

Proof Techniques

B.1.3 Proof Templates

NOTE

The following templates summarize proof methods that have already appeared and will be used throughout the course.

Universal Statements

RECALL AND TOOLS

A universal statement asserts that every object satisfying a property p also satisfies a property q. It can be written as an implication: if an object satisfies p, then it satisfies q.
To prove the statement, take an arbitrary object and assume only that it satisfies p. Show that q follows. Be careful not to impose additional assumptions on the object.
To disprove the statement, find an object satisfying p but not q. One such counterexample is enough to disprove a universal statement.

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EXERCISE B.3

Please prove that

$$\forall x \in \mathbb{R},\ (x > 3) \Rightarrow (x^2 > 9).$$

WORKED SOLUTION

Let $x \in \mathbb{R}$ satisfy $x > 3$. Since x is positive, multiply both sides of the inequality by x without reversing its direction:

$$x \cdot x > 3 \cdot x.$$

Hence

$$x^2 > 3x.$$

Also, $x > 3$ implies $3x > 9$. By transitivity,

$$x^2 > 9.$$

We have therefore shown that

$$(x > 3) \Rightarrow (x^2 > 9).$$

Because x was arbitrary, the conclusion holds universally:

$$\forall x \in \mathbb{R},\ (x > 3) \Rightarrow (x^2 > 9).$$

EXERCISE B.4

Please prove by contradiction: if $A \setminus B = \varnothing$, then $A \subseteq B$.

WORKED SOLUTION

Assume that $A \setminus B = \varnothing$ and, for a contradiction, that $A \not\subseteq B$. Then some $x \in A$ satisfies $x \notin B$, so $x \in A \setminus B$. This contradicts the assumption that the difference is empty. Therefore $A \subseteq B$.

EXERCISE B.5

Please disprove the following statement by a counterexample: $x^3 > x$ for every $x \in \mathbb{R}$.

WORKED SOLUTION

The statement is false. Take $x = \frac{2}{3}$. Then

$$\left(\frac{2}{3}\right)^3 = \frac{8}{27} < \frac{2}{3},$$

so the proposed strict inequality fails.

Created by Chen Gadi

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